0.1 Question 1: Human Context and Ethics

In this part of the project, we will explore the human context of our housing dataset. **You should watch Lecture 13 before attempting this question.**

0.1.1 Question 1a

Consider the following question: *"How much is a house worth?"*

Who might be interested in an answer to this question? Be sure to list at least three different parties (people or organizations) and state whether each one has an interest in seeing a low or high housing price.

Your response should be approximately 3 to 6 sentences.

Homebuyers want to see low prices so they can pay less for the property. Home sellers prefer high prices because they receive more proceeds. Real estate agents prefer high prices because they receive more commission

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0.1.2 Question 1b

Which of the following scenarios strikes you as unfair, and why? You can choose more than one. There is no single right answer, but you must explain your reasoning. Would you consider some of these scenarios more (or less) fair than others? Why?

A. A homeowner whose home is assessed at a higher price than it would sell for.

B. A homeowner whose home is assessed at a lower price than it would sell for.

C. An assessment process that systematically overvalues inexpensive properties and undervalues expensive properties.

D. An assessment process that systematically undervalues inexpensive properties and overvalues expensive properties.

Your response for each chosen scenario should be approximately 2 to 3 sentences.

I choose C and D. Scenarios A and B describe a single home being mis-assessed, which could be random noise, appeals and rechecks can correct those. By contrast, C and D indicate systematic bias, errors that repeat across groups, so they create persistent, unequal tax burdens and erode trust. Between them, D is less unfair to me than C. In D, high-value owners are overtaxed and low-value owners are undertaxed, which resembles a progressive tax structure aligned with ability-to-pay and can modestly reduce wealth inequality. However, D is still not ideal. C is worse because it is regressive, it overtaxes lower-value households while subsidizing wealthier owners, it exacerbating inequality.

0.1.3 Question 1d

What were the central problems with the earlier property tax system in Cook County as reported by the Chicago Tribune? What were the primary causes of these problems?

Your response should be approximately 2 to 4 sentences.

Note: Along with reading the paragraph above, you will need to watch Lecture 13 to answer this question.

The Tribune found that Cook County's assessment system was inaccurate and regressive, it routinely overvalued lower-priced homes and undervalued higher-priced homes, producing racial and neighborhood disparities. Two main causes are: the CCAO relied on a simple linear model which tends to be regressive, and the Board of Review's appeals process disproportionately benefited wealthier homeowners, lowering their tax burdens while raising those of poorer owners.

0.1.4 Question 1e

In addition to being regressive, how did the property tax system in Cook County place a disproportionate tax burden on non-white property owners?

Your response should be approximately 3 to 4 sentences.

Beyond being regressive, the system hit many non-white owners harder. Their homes were often in lower-priced areas that the model over-assessed, so they paid more tax than they should. Wealthier, whiter areas were under-assessed and their owners appealed more often and won, cutting their bills. Many non-white owners faced higher costs and less help to appeal, so the extra burden stayed on them.

0.2 Question 4a

We can assess a model's performance and quality of fit with a plot of the residuals $(y - \hat{y})$ versus the observed outcomes (y).

In the cell below, use `plt.scatter` (documentation) to plot the **model 2** residuals of `Log Sale Price` versus the original `Log Sale Price` values. For this part, you only need to plot the residuals and outcomes for the **validation data**.

- You should also **ensure that the dot size and opacity in the scatter plot are set appropriately** to reduce the impact of overplotting as much as possible. However, with such a large dataset, it is difficult to avoid overplotting entirely.

In [29]:
```
resid_m2 = Y_valid_m2 - Y_predicted_m2
plt.scatter(
    Y_valid_m2,
    resid_m2,
    s=6,
    alpha=0.25,
    edgecolor='none'
)
plt.xlabel('Log Sale Price(Validation)')
plt.ylabel('Residual (y - ŷ) for Model 2')
plt.title('Model 2 Residuals vs. Log Sale Price (Validation)')
plt.show()
```

Model 2 Residuals vs. Log Sale Price (Validation)
Residual (y - ŷ) for Model 2
Log Sale Price(Validation)
2
0
−2
−4
−6
−8
6
8
10
12
14
16

0.2.1 Question 6c

Using the functions above, we can generate visualizations of how the RMSE of sale price and proportion of overestimated houses vary for different intervals:

```
In [ ]: # RMSE plot
        plt.figure(figsize = (8,5))
        plt.subplot(1, 2, 1)
        rmses = []
        for i in np.arange(8, 14, 0.5):
            rmses.append(rmse_interval(preds_df, i, i + 0.5))
        plt.bar(x = np.arange(8.25, 14.25, 0.5), height = rmses, edgecolor = 'black', width = 0.5)
        plt.title('RMSE of Sale Price For Different Intervals\n of Log Sale Price', fontsize = 10)
        plt.xlabel('Log Sale Price')
        plt.yticks(fontsize = 10)
        plt.xticks(fontsize = 10)
        plt.ylabel('RMSE')

        # Overestimation plot
        plt.subplot(1, 2, 2)
        props = []
        for i in np.arange(8, 14, 0.5):
            props.append(prop_overest_interval(preds_df, i, i + 0.5) * 100)
        plt.bar(x = np.arange(8.25, 14.25, 0.5), height = props, edgecolor = 'black', width = 0.5)
        plt.title('Percentage of House Values Overestimated \n for different intervals of Log Sale Pric
        plt.xlabel('Log Sale Price')
        plt.yticks(fontsize = 10)
        plt.xticks(fontsize = 10)
        plt.ylabel('Percentage of House Values\n that were Overestimated (%)')

        plt.tight_layout()
        plt.show()
```

Which of the two plots above would be more useful in ascertaining whether the assessments tended to result in progressive or regressive taxation? Provide a brief explanation to support your choice of plot.

Then, explain whether your chosen plot aligns more closely aligns with scenario C or scenario D from `q1b`:

```
C. An assessment process that systematically overvalues inexpensive properties and undervalues expensiv
D. An assessment process that systematically undervalues inexpensive properties and overvalues expensiv
```

Your response should be approximately 3 to 4 sentences.

Type your answer here, replacing this text.

0.3 Question 7: Evaluating the Model in Context

0.4 Question 7a

When evaluating your model, we used RMSE. In the context of estimating the value of houses, what does the residual mean for an individual homeowner? How does a positive or negative residual affect them in terms of property taxes? Discuss the cases where the residual is positive and negative separately.

Your response should be approximately 2 to 4 sentences.

Type your answer here, replacing this text.

0.5 Question 7b

Reflecting back on your exploration in Questions 6 and 7a, in your own words, what makes a model's predictions of property values for tax assessment purposes "fair"?

This question is open-ended and part of your answer may depend on your specific model; we are looking for thoughtfulness and engagement with the material, not correctness.

Hint: Some guiding questions to reflect on as you answer the question above: What is the relationship between RMSE, accuracy, and fairness as you have defined it? Is a model with a low RMSE necessarily accurate? Is a model with a low RMSE necessarily "fair"? Is there any difference between your answers to the previous two questions? And if so, why?

Your response should be approximate 1 to 2 paragraphs. Feel free to answer the questions in the hint to structure your answer.

Type your answer here, replacing this text.

0.6 Question 8: Finding Better Metrics For Our Model

0.7 Question 8a

As discussed in Project A1, RMSE—while a widely used and powerful error metric introduced in earlier lectures—is not always the most appropriate choice when evaluating the *fairness* of a property appraisal system. In Question 7, you already encountered some of RMSE's limitations, particularly its tendency to disproportionately emphasize errors in high-priced properties due to the squaring of residuals.

In this question, rather than relying on RMSE, we will train and evaluate our model using another custom fairness metric. Specifically, we will examine the **Mean Absolute Percentage Error (MAPE)**, which measures the average error as a percentage of the true value. This allows us to better assess whether the model is making relatively fair predictions across different segments of the housing market.

In the code cells that follow, we'll explore how MAPE varies across price ranges—comparing the model's relative performance on inexpensive versus expensive housing. This helps identify whether the model systematically favors or disadvantages certain groups of properties based on their price.

Note: You'll notice that we're no longer using `lm.LinearRegression()`, but instead have started using `minimize`. This is because `scikit-learn` does not allow customization of the loss function in their standard linear regression model. The approach shown below provides an equivalent way to train a linear regression model using a custom error metric.

Warning: These cells take quite a long time to run. Please be patient, wait, and avoid restarting the kernel or rerunning these cells more than necessary.

In [ ]:
```
data = pd.read_csv("cook_county_train.csv", index_col='Unnamed: 0')
trainX, trainY = feature_engine_final(data)
```

In [ ]:
```
# X is the design matrix (including bias column), y is the vector of true outputs, theta is the
def mape(theta, X, y):
    y_pred = X @ theta  # compute predicted values using linear combination of features
    percentage_error = np.abs((y - y_pred) / y)  # calculate element-wise percentage errors
    return np.mean(percentage_error)  # return the MAPE
```

In [ ]:
```
from scipy.optimize import minimize

# Add bias (intercept) column to the design matrix
trainX_with_bias = np.column_stack([np.ones(trainX.shape[0]), trainX])

# Initialize parameter vector with zeros
```

```
theta_0 = np.zeros(trainX_with_bias.shape[1])

# Use scipy's minimize to find weights that minimize MAPE
res = minimize(mape, theta_0, args=(trainX_with_bias, trainY), method='BFGS')

# Optimal weights after training
theta_opt = res.x

theta_opt
```

In [ ]:
```
new_preds_df = pd.DataFrame({
    'True Log Sale Price'     : trainY,
    'Predicted Log Sale Price': trainX_with_bias @ theta_opt,
    'True Sale Price'         : np.e ** trainY,
    'Predicted Sale Price'    : np.e ** (trainX_with_bias @ theta_opt)
})

plt.figure(figsize=(8, 5))
plt.subplot(1, 2, 1)

mape_values = []
for i in np.arange(8, 14, 0.5):
    mape_values.append(mape_interval(new_preds_df, i, i + 0.5))

plt.bar(x=np.arange(8.25, 14.25, 0.5), height=mape_values, edgecolor='black', width=0.5)
plt.title('MAPE of Sale Price Across\n Log Sale Price Intervals', fontsize=10)
plt.xlabel('Log Sale Price')
plt.ylabel('Mean Absolute Percentage Error (MAPE)')
plt.xticks(fontsize=10)
plt.yticks(fontsize=10);
```

What can you infer from this graph using our new custom error metric? Write your findings in the space below in three to five sentences. Consider addressing the following points:

- How does this metric differ from RMSE?
- What is the purpose of this custom error function?
- How does it relate to the idea of underpriced expensive housing mentioned in Question 6a?
- Why would this potentially be better than the RMSE in terms of the CCAO dataset?

Type your answer here, replacing this text.